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ENVS 397: Senior Capstone in Environmental Studies

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Saturday May 9th, 2026

Water Quality Curriculum for Germantown based 5th and 6th Graders

Executive Summary

The development of a water quality curriculum for 5th and 6th graders emerged from our collaboration with The Water Shed, a community oriented organization in Germantown, Philadelphia working to educate and empower flood resilience through art, education, and science. Our work is inspired by the educational opportunities that The Water Shed has offered along with the meaningful learning opportunities we were able to experience when visiting their building in Germantown. Local education and outreach is essential for creating a strong community that can effectively organize, advocate for their needs, and seek environmental justice. The issues of flooding and water quality are pressing in Germantown and Philadelphia as a whole, connected to issues of health, infrastructure, poverty, climate change, legislation and more. The goal of this project was to create a series of lesson plans that can help to educate a new generation of informed citizens to feel confident in analyzing issues intersectionally, take initiative, work with others, and to have fun learning. The following paper includes a background about the types of learning we incorporated into our lesson plans—verbal, visual, interactive—and why they are important, our reasoning for our specific type of curriculum, a background on the content of the lessons that can be referenced by teachers, and finally, the outline of our water quality curriculum.

Introduction

Urban flash flooding is unfortunately all too common in the neighborhood of Germantown. Because of the almost two century old underground sewage infrastructure in the area, when there is heavy rain it overwhelms the system and spills out into the streets, damaging homes and businesses and even leading to death. In September of 2011, a woman became trapped in her car due to the floodwaters and passed away, a devastating loss that occurred due to its placement in the historic Wingohocking watershed, its outdated infrastructure, and the exacerbating effects of climate change. These issues of flooding have been at the forefront of the residents' minds and there has been a lot of action to take a community focused approach to flood mitigation and resilience. The City of Philadelphia created a Germantown Community Flood Risk Management Task Force which has sought to engage the community in "neighborhood-based stakeholder coalition" and there has been a lot of engagement from the Water Department to provide educational materials in collaboration with community stakeholders (Maiorano). While there has been a lot of educational efforts put into addressing flood resilience, which is the most pressing problem for Germantown. However, many environmental justice issues are connected and the issues of water quality we chose to focus on stem from the same source: the outdated sewage infrastructure and climate change.

Additionally, as climate change gets worse and temperatures rise, the atmosphere will increase its capacity to hold moisture, leading to more intense storms. Given the already aging infrastructure of Germantown houses, flooding, and therefore water quality, could worsen if the root causes are not addressed. Education has been essential for providing community members with the tools to advocate for flooding solutions and because of this, we took the same approach with our water quality curriculum. It was also important for us to target younger students with our curriculum so that they are given the tools at a young age to grow into capable adults and advocates, should they want to be.

Water quality is also a way for students to be introduced to environmental justice issues from a context in which they can relate to. Environmental justice, as defined by Miles et al., "calls attention to the ways that environmental harms—including hazardous waste, land use and extraction, and toxins released in the production of capital—are inequitably distributed within marginalized communities." Water is an essential resource on this planet and according to the United Nations, over two billion people didn't have access to safe drinking water in 2022 and "roughly half of the world's population is experiencing severe water scarcity for at least part of the year." Access to clean water is a health and environmental justice issue that affects people no matter how old they are and is therefore important to learn about. By connecting environmental issues with health problems, infrastructural issues, and the effects of poverty, students will learn to think in an interdisciplinary manner. Interdisciplinary thinking is necessary for understanding the world as an interconnected system.

Learning Approaches

There are many different types of learners and modes of learning that must be integrated into a curriculum in order to create an educational program that presents every student with the opportunity to engage and to succeed. The following three learning approaches—interactive learning, visual learning, and verbal learning—were taken into account and incorporated into our curriculum.

<u>Interactive Learning</u>

Interactive learning is “a student-centered approach to teaching and learning that incorporates hands-on activities, collaboration, discussion, and technology support.” This approach allows students to be in charge of their own learning while still having a teacher there to monitor them and provide them with immediate feedback. For this type of learning to work, students need to be presented the opportunity to learn through interacting with the environment rather than simply listening to someone else explain a topic. It has been shown that when students are more engaged in the learning process, they are able to better retain the information and are willing to learn more. The goal of this lesson plan is not just for students to learn about water quality, but also help them learn to problem solve and come up with ideas and solutions to real world issues.

There are many benefits to this learning format which stems from higher levels of attention, engagement, satisfaction with learning, creativity, critical thinking, and retention. Interactive learning is very accommodating and can easily be shifted based on the support needs of different students as well as across different subjects and courses. Additionally, the style introduces students to collaboration techniques that they will need for future careers while also helping them understand how their work as students connects back to the real world. Lastly, and potentially most importantly, students are able to build confidence in their knowledge and skills through self-efficacy and completing tasks on their own.

Although the focus of interactive learning is to allow the students to take the lead when it comes to the project, there still needs to be some structure. First, you need a good starting point to gauge where students are in their learning so that there is a better understanding of who needs more help and who already has a better grasp of the information. By understanding the different skill levels, the teacher is able to ensure that the project is one that can be shifted depending on varying support needs so that no student feels overwhelmed or overly confused. Second, there needs to be some form of a vague outline for the students' learning so that they are still learning on their own yet have some guidance to lead themselves with. Additionally, there should be set goals and provided readings for them to use and learn from so that they can find the answers to their questions through hands-on activities rather than just lectures. Regardless of whether or not the students are at the center of the learning, there needs to be an open dialogue between them and the teacher so that they feel like they are in control of their own learning without feeling completely lost. Third, there needs to be some form of rubric so that the students can progress their skills while having the ability to reflect on how they have improved. This rubric should also incorporate personalized goals so that the students feel motivated to learn and pay attention to and apply feedback.

Visual Learning

Visual learning has a place in all aspects of learning due to its ability to be adapted to all age ranges, cultural backgrounds, and educational systems. Visual learning is "a teaching and learning approach that emphasizes the use of visual aids to enhance understanding and retention of information", helping students process information through seeing it played out in front of them. Visual learning allows for new concepts to be better integrated since the students have something concrete that they can connect the information to. The process involves the student organizing the idea that they are presented with, processing the information, and then prioritizing the important aspects of the overall information. This learning style centers more around the brain and mental retention aspect of learning rather than the literal action of acquiring the information.

There are many benefits to visual learning that help students progress and develop good habits. Students are able to enhance their memory and critical thinking through their ability to call upon information that they have seen and then use that information to interpret similar situations. Additionally, visual learning has the ability to foster inclusivity through a wide media usage that can be adapted to suit different preferences and thus reach a broader audience.

While visual learning works for a wide range of students, there are certain types of learners who work best with this format, with the main type of learner being those who are structure-based, who learn best when provided with a concrete example of the information. This could mainly be due to the ability of visual stimuli to help with organizing and visualizing the information, providing something to call back on directly rather than having to simply memorize the information. With the information being cemented in the student's mind, there is room for further thought and discussion amongst students when it comes to how they interpret the information. All of this is to show that the goal of this approach goes beyond just learning the information, it is about conceptualizing it so that it can be fully understood and applied later.

Verbal Learning

Verbal learning is an approach that is centered around language. The learning style involves "the acquisition, retention and retrieval of information that is denoted in the form of words". Through being a details based learning style, verbal learning is able to have connections to a wide range of learning goals, such as language comprehension, knowledge acquisition, and knowledge retention as well as provides ample opportunities for group work development. Visual learning combines personality, cognition, motivation, and material importance to create a well-rounded approach to learning. Even general exposure to verbal content can result in improvement in general learning, helping the students' to expand their vocabulary and reading skills.

There are 4 main types of verbal learning, serial learning, paired associate learning, free recall learning, and recognition learning. Each of these is unique and thus are each suited for different types of learners and information. Serial learning is based around sequential memorization and is used when the order of the information being learned is important. Paired associate learning involves paired memorization where both parts of the information must be put together for the overall information to make sense. This allows the student to be able to recall one half of the information when they are presented with the other half, improving concept connection and building long-term memory. Free recall learning is the opposite of serial learning where the order is irrelevant and the main focus is storing the information. Lastly, recognition learning mainly involves differentiation between correct and incorrect information with the main focus being identification. Each of these types has importance in the recall aspect of visual learning and with all of them working together, language development is much easier for students.

Language is at the center of almost all learning, without it there would be no ability to understand or process concepts. Two major characteristics of verbal learners that are based off of language are vocabulary and memory, with each being highly intertwined. Good vocabulary allows students the ability to remember, interpret, and apply new terms outside of a learning environment. Once the process is already engrained in the students' brains, it is easier for them to apply new terms outside of a learning environment as well as learn new vocabulary which can then be remembered and interpreted. Students will be able to comprehend more complicated concepts and will have more ability to express their opinions and interpretations on the subject.

With students gaining all of this new information, they need to be able to have good memory, which is improved through students being able to process more information at a faster pace which allows for them to get through more lessons.

As stated, memory is imperative to the learning process and without the ability to retain knowledge learned, students would not be able to advance further in their learning. In addition to this obvious benefit, retention is also a crucial part of maintaining a high writing level. By having improved writing abilities, students are able to better convey ideas and organize their thoughts as well as combine their own feelings on a topic with the information that they have learned. These improved abilities also help students with confidence in their knowledge due to the fact that they are remembering and applying it. To get to this point though, students must be able to acquire the knowledge in the first place before they are able to maintain and apply it.

Teachers are able to use the format of developing language and knowledge followed by retention, provided by verbal learning, to help develop strategies to make the information more meaningful and effective. Acquiring knowledge is possibly the most important aspect of this approach when it comes to studying academic concepts. In relation to verbal learning, it applies greatly to the improvement of student reading skills. Students have the ability to learn from a wide range of fields which they can then apply later to other readings, thus providing them with the ability to compare and contrast the information that they are gaining. With the acquired knowledge, students form and take in different perspectives while still being able to articulate their own point.

Methods

For our lesson plans, we chose to focus on 5th and 6th grade aged children. We decided on this so that the children would be young enough for the relatively activity based learning format, yet old enough to conduct the water quality tests on their own, comprehend the information and case studies, and apply the information to real life. We formatted the lesson plans into smaller adaptable lessons that are able to be exchanged, reorganized, or even done as standalone lessons/activities. We wanted to make the lessons as accessible as possible so that they can be utilized for a variety of different school and classroom setups. Additionally, we wanted to make sure that the lessons could even be conducted in a summer camp or after school program setting. By making the lessons smaller, both teachers and non-teacher leaders are able to decide what works best for their class or group. We created 10 small segments of a lesson, each equipped with background information, and sources that can be combined in whatever way the teacher sees fit. We were able to combine aspects of interactive, visual, and verbal learning approaches in our plans through the use of hands-on activities and groupwork, maps and videos, as well as presentations and readings. Additionally, we varied the combinations of approaches with some lessons mainly utilizing one approach while others use multiple. We wanted to ensure that the lessons were as open-ended as possible while still providing structure and were able to achieve that through our planning process and wide range of source material.

While our lesson plans about water quality issues in Germantown and Philadelphia are made to be adaptable for any educational context, they are also able to fit into the public school curriculum of Philadelphia. According to the School District of Philadelphia's Parent Guides, in 5th grade students will learn in science class about how much water is found in different places on earth and how matter cycles through the ecosystem. 6th graders will learn about the factors that influence weather and how we know that climate change is happening. At each level, students will have a very basic background in the water cycle, the availability and importance of water, and climate change. Our water quality adds onto this by connecting these two topics. For 6th graders this will mean connecting back to a previous year of learning to understand how education is constantly building on itself.

Content Background

Our lesson plans aim to educate about water quality in Germantown and the greater Philadelphia area in multiple ways. First is a historical approach where students can connect present day issues with their root causes from the past. Second is an experiential learning approach where students can take water quality testing into their own hands, showing that they have the capability and agency to take these issues into their own hands. The water testing combines a field trip, groupwork, and data collection along with an activity to practice graphing and analyzing data. Last is a look at case studies and a brainstorming session to apply what the students have learned to contexts beyond Germantown and Philadelphia and to create a sense of hope and possibility. The following sections provide background research that is relevant for the history and water testing portions of the curriculum.

History of the Sewage System

The many issues with water quality that Philadelphia faces today can be traced back to the creation of the city's combined sewer system that transports stormwater, sewage and the original flow of the city's historical creeks together in the same pipes. While there are other sources of pollution such as agricultural runoff, oil spills, and trash/waste that got thrown in storm drains, we decided to focus our lesson plans on the issues of sewage overflow and chemical contamination from old piping. Through these two forms of contamination, the students could connect current water quality issues to the creation of Philadelphia's sewer system and the importance of water in the city's history.

Philadelphia was an important port city built along the Delaware River on top of historic wetlands and mudflats. The city's area consisted of eight watersheds: Delaware, Schuylkill, Wissahickon, Tookany/Tacony Frankford, Pennypack, Poquessing, Cobbs, and Darby. Before European settlement, the city of Philadelphia contained a complex creek system which has since been subsumed underground (Figure 1) (The Water Center). The historic streams were central to the establishment of industry in the city. Philadelphia was the home to the papermaking industry which made use of the region's waterways to power paper mills. While the papermaking industry proliferated in the 19th century, by the early 21st century most of the mills had closed. Along with papermaking, Philadelphia was a large textile manufacturing center that emerged from the Industrial Revolution. The textile industry grew fast from twelve thousand workers in 1850 to sixty thousand workers in 1882 (The Encyclopedia of Greater Philadelphia). Overall, the rivers were key to this growth in industry, supplying power to a variety of mills.

While the economic success of Philadelphia relied heavily on the waterways, this same industrial growth attracted more and more people to the city causing a deterioration of sanitary conditions. During the 19th and 20th centuries, human and industrial waste was dumped straight into the waterways and carried off by gravity. However the pollution and concentration of bacteria from the waste led to widespread sickness and disease. A closer look at the sanitation system at the time reveals the impetus for the modern day combined sewage system and shows how closely tied health and water have been since the beginning of the city's history. During the first half of the 19th century, people would have brick lined outhouses called privy pits in their backyards that would fill up and release strong odors. At the time, strong odor was believed to cause disease and in order to deal with the stink of the privy pits, the Philadelphia Board of Health required homeowners to have them emptied. The waste was picked up by “night men” and dumped into “poudrette pits” on the outskirts of the city which in turn would mix with charcoal, swamp muck, gypsum and other materials to decompose and then be used as fertilizer. When flush toilets became popular in the second half of the 19th century, they would quickly overwhelm privy pits. The overtaxing of the privy pits led to the creation of a citywide sewage system that connected houses and businesses to nearby bodies of water through piping. The dumping of human waste into the rivers efficiently delivered bacteria and other disease causing organisms straight into the drinking water system. Diseases ran rampant, the most common being typhoid fever. More than 27,000 people died from typhoid fever between 1860 and 1909 when the Board of Health published cause of death data. Finally, in 1909, the city built a water filtration system (Water History PHL).

In order to create the sewage system, the city’s engineers laid piping in the city’s natural streambeds. This strategy made use of gravity to carry the mix of sewage and streamwater through the piping system and also saved money and labor by following the pre-made channels that the water had carved out over centuries. This new sewage system subsumed 200 miles of Philadelphia streams and would carry the original stream flow, sewage from homes and businesses and stormwater runoff from the streets. As the city subsumed the original system of creeks into the sewage pipes, they also filled in the land above, evening out the city’s landscape. The flattening allowed for efficient transport and the ability to create the grid system that exists today. In some locations such as Logan, the floodplains were filled up to 40 ft (TTF Watershed).

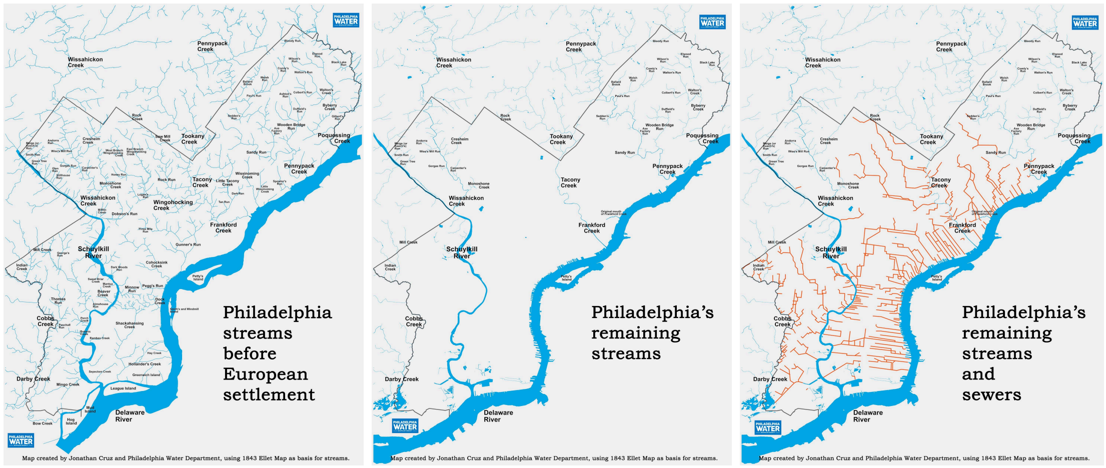

Figure 1: Philadelphia's streams before European settlement, the streams today, and the current day sewers. Image from Water History PHL

Germantown, where The Water Shed is located, is in the Tookany/Tacony-Frankford Watershed. The historic Wingohocking creek used to run through the region, making up a third of the watershed. This creek was completely subsumed in the late 19th century with the process lasting a staggering 40 years. The Wingohocking was one of the creeks with the greatest transformation when the sewer system was created and no longer exists on modern maps. In 2015, the TTF Watershed Partnership—a hands-on education, stewardship, restoration, and advocacy group—hosted a tour of the Wingohocking creek. The tour followed the evidence that remained in the city's landscape of its existence and also stopping at the one spot where the creek had been daylit in Awbury Arboretum where a small pond and a trickle of stream flow can be seen. Some of the winding streets in Logan and Germantown reflect the pathway of the historic Wingohocking creek and there are many historic mill buildings that were once on the river's banks. The stops on the tour that have been marked on their map are also great spots for teachers to take 5th/6th graders to in order to learn more about this history and apply what they learn about water quality to test the daylighted water flow (Figure 2).

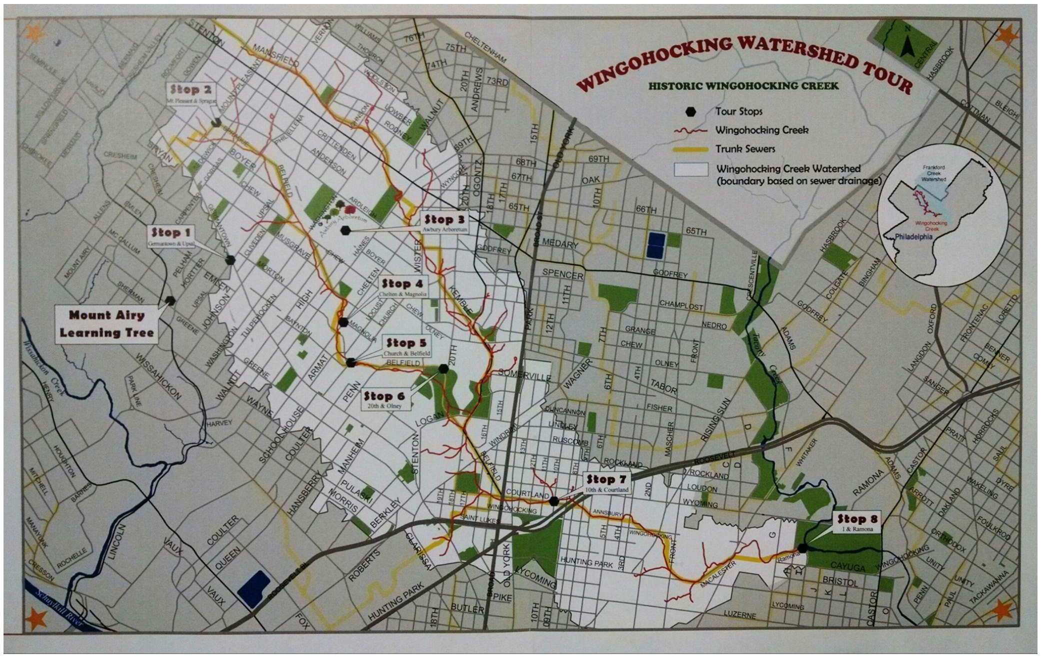

Figure 2: stops on the Wingohocking Watershed Tour. Image from TTF Watershed

The same system that is the cause of Germantown's flooding issues is also the reason for Philadelphia's main water quality issue: combined sewage system. The fact that Germantown was built on the old watershed and floodplains of the Wingohocking is a major cause for flooding today, aided by the inefficient combined sewer system. A combined sewer system, as mentioned earlier, is a system where both stormwater and wastewater flow through the same pipes underground. Around 60% of Philadelphia is served by this combined sewage system and the other 40% uses separate pipes for sewage and stormwater (Penn Environment). A huge problem with the combined sewage system is that it leads to combined sewage overflows (CSOs), contaminating the rivers. In dry weather, stormwater and wastewater combine in the pipes and are all carried to the wastewater treatment plant. However, in heavy rain, the system fills to capacity and excess, untreated water is released into the Tacony-Frankford creek through outfall pipes. According to Penn Environment's report on sewage pollution in the

Philadelphia-Camden region published in 2025, 12.7 billion gallons of raw sewage mixed with polluted stormwater was dumped into local waterways between 2016 to 2024 due to CSOs. Climate change will also significantly exacerbate this issue, leading to increased precipitation intensity which in turn is predicted to increase annual volume of overflows and slightly increase the number of annual CSO events (Hensyl et al.).

In 2011, Philadelphia committed to a 25 year plan to reduce pollution and restore local waterways called Green City, Clean Waters. The goal of this plan was to use green infrastructure such as rain gardens, green roofs, and permeable surfaces along with engineering methods such as underground holding tanks to soak up stormwater and slow down its entry into the sewage system, preventing overflow situations. Reports from the water department show that the city has been meeting all of its targets and there has been a 21% improvement from over a decade ago in overflow reduction. At the same time, critics of the city's strategy point out that this improvement is less impactful when increased rainfall due to climate change is taken into account (Bagenstose). This Green City, Clean Waters plan has made improvements but there is still a need to target the root of the issue which exists in the city's two century old sewer system and improve family home infrastructure in an affordable manner.

Bacteria

While pollution from combined sewage overflow never reaches our drinking water, it has major impacts on the ecosystem and human health. According to the Delaware Health and Social Services, the bacteria, fungi, parasites and viruses that can be found in sewage and wastewater can cause lung and intestinal infections and can worsen allergies and other illnesses. This can be dangerous for recreation in the waterways such as swimming, boating, and fishing. Common bacteria and diseases present in wastewater are E.coli, shigellosis, typhoid fever, salmonella, and cholera. Recreation in polluted waterways less than 24 hours after a CSO event can increase risk of acute gastrointestinal illness by 39 to 75% compared to any other time of the year (McGinnis et al.). Along with adverse human health effects, sewage pollution is dangerous for the aquatic wildlife because it can cause algal blooms and deplete the oxygen in the water—consequently killing fish and other wildlife. Due to the proliferation of bacteria in local waterways, especially after storms, our lesson plans will have students test for bacteria as part of their hands-on water quality testing activity. Bacteria is an easily testable contaminant that is included in many affordable at home testing kits.

Lead and Copper

Along with bacteria, two other simple but extremely important contaminants are lead and copper. These are the two main chemicals of concern when it comes to water quality in Germantown. Unlike bacteria, lead and copper do affect drinking water because of their presence in outdated home plumbing systems. Lead is a toxic metal while copper is an essential micronutrient that can cause problems when ingested in high doses. These chemicals are also both drinking water corrosion compounds that are particularly concerning in Germantown because of its outdated plumbing infrastructure. Unfortunately lead and copper are used in older service lines and household plumbing systems due to their flexibility and thus anti-corrosion materials are needed to combat the potential of these chemicals being released given the extreme harm that these chemicals can cause if ingested.

There are many ways in which these chemicals can cause harm to individuals, especially children. For instance, lead can be harmful even at low levels and can build up in the body over time. While the symptoms vary between adults and children as well as between levels, there are some overarching issues that can be experienced by everyone such as problems with bones, teeth, blood, liver, kidneys, and the brain. Ingesting lead can also lead to lead poisoning which can cause symptoms such as high blood pressure, cardiovascular disease, joint and muscle pain, difficulty with memory or concentration, and reproductive issues in adults as well as irritability, weight loss, abdominal pain, fatigue, vomiting, and seizures in children. Additionally, there is also variability when it comes to the level of symptoms. Low level symptoms involve interference with thought processes, lower IQ in children, and attention and behavioral problems while moderate level symptoms involve hearing loss, anemia, hypertension, kidney impairment, immune system dysfunction, and toxicity to reproductive organs. Unlike lead, while copper ingestion can still be harmful, it is truly only concerning if it is ingested in larger doses. Too much copper ingestion can lead to symptoms of gastrointestinal tract issues if the exposure is acute and cause short term nausea as well as affect the liver and kidneys if the exposure is long-term. Additional symptoms include headaches, vomiting, diarrhea, stomach cramps, nausea, liver damage, kidney disease, and red blood cell damage. Lastly, there are issues specific to men, such as fertility issues, and can be much more harmful to people who are more sensitive, such as those with Wilson's disease and infants under 1 year old. All of these health concerns are major reasons as to why each of these chemicals must be focused on within water quality learning.

Lesson Plans

The following plan is structured as a compilation of short lesson plans that are flexible and adaptable. They can be incorporated into a class curriculum, led as an educational program at The Water Shed, or as part of an afterschool program. The activities provided are meant to be on the shorter side and multiple activities may be combined together to form a class.

History

Trip to The Water Shed

Students can either go to The Water Shed to conduct any of the lessons or simply use this as an opportunity to interact with the community and learn about the water system directly from people living and working in the area. Students can also come prepared with questions to ask or the teacher/leader can go through the website with the students or provide handouts with background information on what The Water Shed is and what they do. If The Water Shed is able to present about the history of Philadelphia's creeks and sewer system then the next presentation activity can be skipped and replaced with a class reflection.

Materials and Resources

- The Water Shed → https://www.water-shed.org/

Presentation

This presentation is centered around the history behind water and water systems, both in Germantown and in general. The presentation can cover the information provided in the History of the Sewage System, Bacteria, and Lead and Copper sections of this paper. The object of this presentation should be to connect Germantown and Philadelphia's history with the current issues of combined sewage overflow and lead/copper contamination.

Materials and Resources

Optional: Before the presentation, provide students with a list of important vocabulary they should be able to define by the end. At the end of the presentation, construct definitions for each of the terms as a class. Suggested vocabulary (select what makes sense for your class):

- Combined Sewer System
- Combined Sewer Overflow (CSO)
- Runoff
- Sewage:
- Wingohocking Creek
- Watershed
- Sewer outfall
- Water mill
- Bacteria
- Climate change
- Precipitation
- Fill
- Algal bloom/Algae
- Lead
- Copper
- Corrosion

In replacement of, or in addition to, the presentation, the teacher can also play this video where the archivist Adam Levine summarizes the history of Wingohocking creek's containment and creation of the sewer system. The benefit of this video is that it combines speaking with great visuals.

Map Activity

This activity allows students to connect their learning to themselves in order to help cement the information that they are learning in reality so that they can feel closer connected to the community. Students will examine a map of historical creeks and will be asked to find their home or a place they frequent on the map. They will be asked how far away their chosen location is from the historic Wingohocking Creek and what they can infer from that. Is their chosen location on a hill or in a valley? Where might the valleys have existed in the past based on the map? If desired, the teacher can also show other images and the students can relate them back to the previous presentation.

Materials and Resources

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  Sourced from [TTF Watershed]()

- [Slideshow]() of more images and maps relating to the previous presentation on the history of the sewage system.

<u>*Timeline Activity*</u>

The timeline activity provides a way for students to contextualize the history of Philadelphia's sewer system and water management within the broader history of Philadelphia and connect this history to the present day. For the activity the students will be split into small groups and receive slips of paper with undated events from Germantown and Philadelphia history. The students will then be asked to order the events in chronological order. This activity can occur after the history presentation as a form of knowledge retention or it can be its own separate activity to have students apply their deductive reasoning skills.

Materials and Resources

- The Water History PHL website is ripe with archived historical documents and summaries

Preliminary Testing

Presentation

This presentation is mainly about connecting water contaminants to personal and community health. Teachers can use the information reviewed in the *Lead and Copper* section to compile a summary for the students.

Materials and Resources

- Copper and lead section of our paper

School Water Testing Activity: Scavenger Hunt Part 1

This activity is meant to help students apply the information that they have gained about water contaminants through actual water testing. Prior to the lesson, the teacher should customize the Data Collection Worksheet provided below depending on what contaminants will be tested and what kind of locations the students have access to. The activity starts with a water quality testing kit demonstration that will vary based on the kit that the teacher, school, or leader decides to use (testing kit options are provided below in the *Sources and Resources* section and will have instructions provided by the kit). This demonstration should include:

1. The teacher demonstrates how to use the test strips to test for bacteria, lead and copper. Bacteria will usually require a 48 hour wait so this step can be skipped or you can choose to demonstrate how to collect a sample and the students can collect samples to look at 48 hours later.
2. Divide students into small groups of 3 or 4.
3. Hand out data collection worksheets to groups (provided below).
4. Explain the units of measurement that the specific contaminant is measured in and have the students write the units of measurement in the worksheet.
5. Have each group do a test run for each contaminant using bottled water samples.
6. Give verbal instructions for the scavenger hunt and send the students off to complete the activity.

During the scavenger hunt, the students will test different school water sources for lead, copper, and bacteria. The students are given a list of water location types within the school, camp, etc. and must find and conduct water quality tests for each location. Some location examples are a bathroom sink, a water fountain, and some type of filtered water (such as a Brita). Each of these locations is exchangeable and the list can be adapted to best fit the program conducting the lesson. Have the students fill out their worksheets and return to the classroom/home base.

Materials and Resources

- Data collection worksheet: This worksheet is editable depending on what testing kit is purchased. Some kits test for "bacteria," some test for "E.coli and coliform" and some test for "total coliform bacteria. Make a copy of the worksheet and edit however desired or simply download and use.

    - It is suggested that teachers fill in locations in the left column to guide students. Here are some ideas: bathroom sink, a water fountain, and bottled water.
- Testing kit options
    - Home Depot
    - Amazon Opt 1
    - Amazon Opt 2

Secondary Testing

Creek/River Water Testing: Scavenger Hunt Part 1

In order to provide students with the opportunity to work on comparing and contrasting, we want them to be able to test a wide range of water sources. This trip will involve taking a trip to a nearby body of water to test the water quality. This could mean going to a river or finding the daylighted segment of the Wingohocking in Awbury Arboretum or, if it has recently flooded, students can test the flood water (carefully). Students will join the same group as they were in for *School Water Testing Activity*. Additionally, programs may have students test additional water sources that are nearby, such as a public bathroom sink or a public water fountain so that they have more data to compare.

Materials and Resources

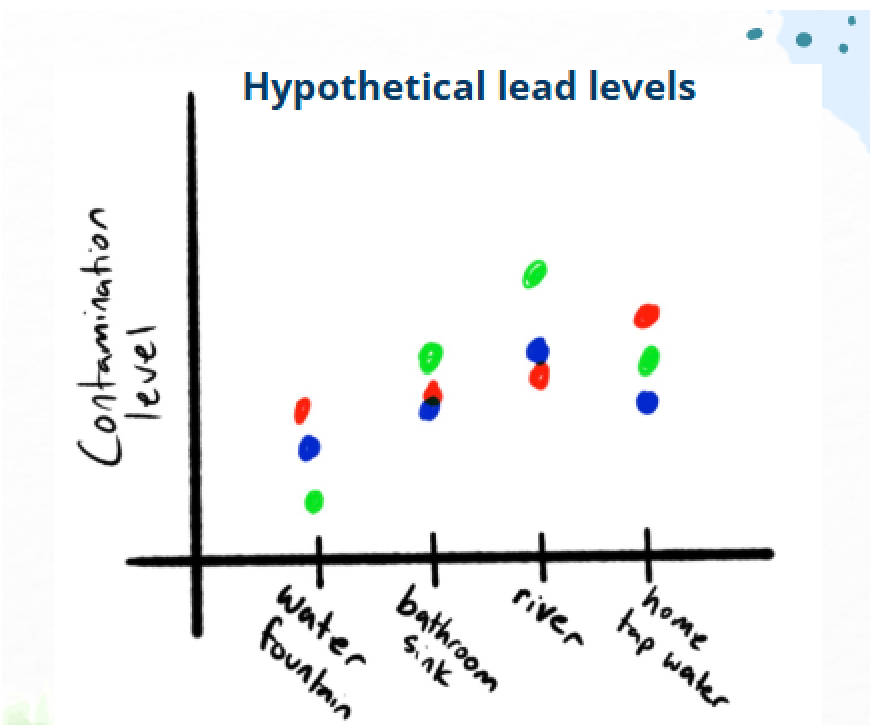
- Data collection worksheet: Same worksheet as above. This worksheet is editable depending on what testing kit is purchased. Some kits test for "bacteria," some test for "E.coli and coliform" and some test for "total coliform bacteria. Make a copy of the worksheet and edit however desired or simply download and use.

Graphing Activity

To help the students work on visualizing and analyzing data, groups will all draw their data on a communal class poster to compare the many different locations and types of water they have tested.

Preparation:

- The teacher will create a plot outline (either on paper, whiteboard, chalkboard, whatever is available). The teacher will draw a plot for each contaminant that the class has tested for using the image below as a sample layout for each plot (keep in mind that the data points in this graph are random and are only there to serve as a visual aid for creating the plot). The Y-axis will indicate contamination level in ppm (or whatever unit of measurement was used for the specific testing kit purchased) and the X-axis will have a list of the locations from which the students tested water.



Activity:

- Each water testing group will be given a unique color marker (or chosen writing device).
- Each group will have a representative come to the board and draw in their data points into the chart.
- The class will then analyze the plot:
    - First, what kind of plot is this? It is a categorical plot, similar to a scatter plot but the x axis has distinct values rather than a continuous scale of numbers.
    - Are you able to see a pattern on the graph or do the data points look more scattered?
    - What point represents the highest contamination level? What location does it come from? What point represents the lowest contamination level? What location does it come from?
    - Can we draw any inferences from this plot?

<u>Solution Brainstorming</u>

<u>Presentation/Discussion/Case Studies</u>

The main objective of this activity is to have students apply what they have learned to case studies or news articles about water quality beyond just Germantown. The collection of case studies could be success stories, examples of innovation, or accounts of water quality issues depending on the teacher's goal. The teacher may first give a presentation on possible solutions to the water quality crisis in Philadelphia (highlight green solutions) if they desire or can jump right into case studies. The students will each be given a segment of 1 of 3 news articles or web article to read. They will then find the other students who read their same case study and will be asked to discuss 3 main questions:

- What is the problem? Who does it affect? What is a possible solution?

Materials and Resources

These suggested news articles and case studies may be difficult to understand. Teachers can read them and select shorter portions for students to read.

- Possible case studies:
  - South Coast scientists: To improve bay health, start at the source
  - Farmers Keeping Nutrients on the Field, Out of Streams
  - Water as a Human Right: How Philadelphia Is Preventing Shut-Offs and Ensuring Affordability
  - 10 years after Flint, the fight to replace lead pipes across the U.S. continues
  - Echoing Newark: How American Cities Can Replicate Newark's Success in Replacing Over 23,000 Lead Pipes in Under Three Years
  - Greening the grey in Washington DC – a resilience success story

Brainstorming Activity

This activity will be based around scenarios provided in the previous *Sources and Resources* section and is meant to help students with retention. The class will be broken up into small groups that can either be the same as the ones from the *School Water Testing Activity* or entirely new groups. Each group will discuss potential solutions to the given scenario and write them down on Sticky Notes. The class will then come back together and each group will put a Sticky Note on the board, one at a time, and connect it to one of the Sticky Notes already on the board in some way. The final product will be a thought web that the class will then discuss together.

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